Delivery Performance, Delay Risk, and Logistics Efficiency Analysis in Global Supply Chain Operations

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An Exploratory Dashboard-Based Case Study Using Power BI

Research paper

Prepared using the project's cleaned dataset and Power BI dashboard results. The paper reports descriptive findings and does not claim official endorsement by APL Logistics.

Abstract

Reliable delivery performance is central to logistics service quality, customer satisfaction, and operational planning. This paper presents an exploratory, dashboard-based analysis of delivery performance, late-delivery risk, shipping-mode efficiency, and regional variation in a logistics dataset treated in this project as an APL Logistics case study. After data cleaning, the analytical dataset contained 180,517 order records. The analysis used delivery status, actual and scheduled shipping duration, late-delivery-risk labels, shipping mode, customer segment, region, and market. Data preparation and measures were implemented using data-cleaning workflows and Power BI calculations, then presented through four interactive dashboard views. The dataset-level on-time delivery rate was 42.72%, while 57.28% of records were classified as delayed. The late-delivery-risk flag was positive for 54.83% of records. These measures are related but not interchangeable: 98,976 records had a positive risk flag, whereas 103,399 records were classified as delayed. Shipping-mode comparisons showed a positive risk rate of 95.32% for First Class and 38.07% for Standard Class. Among the regional indices displayed, Central Asia had the highest reported Regional Delay Index at 1.68 days, followed by Eastern Asia at 1.65 days. The results indicate where further operational investigation may be useful, but they do not establish causes of delay or demonstrate that changing a shipping mode would improve performance. The paper contributes a transparent KPI framework and an interactive reporting approach for descriptive logistics analysis. Limitations include the absence of a date field for time-series analysis, uncertainty about the dataset's provenance, and the descriptive—not predictive or causal—nature of the methods.

Keywords

Supply chain analytics; logistics performance; delivery delay; late-delivery risk; shipping mode; business intelligence; Power BI; KPI dashboard.

1. Introduction

Logistics operations generate records that can be used to examine delivery outcomes, promised-versus-actual shipping duration, and differences across transport options and geographic areas. However, a large table of shipment records is not always easy for decision-makers to interpret. A structured set of key performance indicators (KPIs), supported by interactive visualizations, can help users move from a high-level overview to more focused comparisons.

This project examines delivery performance and delay-related patterns through a Power BI dashboard. It is framed as an APL Logistics case study because that is the context assigned to the project; the analysis does not independently verify that the underlying dataset was collected from APL Logistics or represents its current operations. The goal is to demonstrate a repeatable descriptive-analytics workflow rather than to make claims about the company's actual service level.

The study addresses four questions: (1) What proportion of orders are early, on time, or delayed? (2) How does the dataset's late-delivery-risk flag vary across shipping modes and customer segments? (3) How do actual and scheduled shipping durations compare by shipping mode? and (4) Which regions show comparatively high values in the project's delay index?

The contribution is practical and methodological: it documents data cleaning, defines the KPIs used, and reports the results in a way that can be audited and reproduced. The dashboard is intended to support exploratory review and prioritization. It is not a predictive model, a causal evaluation, or proof that a specific intervention will reduce delays.

2. Literature Review

Research on supply-chain analytics emphasizes that operational data become useful when combined with domain understanding and suitable quantitative methods. Waller and Fawcett (2013) discuss the role of data science, predictive analytics, and large datasets in supply-chain design and management, while highlighting the importance of analytical skills and supply-chain knowledge. Their work supports the general rationale for connecting shipment-level data with operational questions, but it does not validate the specific dataset or results in this paper.

Research on business intelligence and analytics also positions data-driven methods as a means of improving organizational decision-making. Chen, Chiang, and Storey (2012) review business intelligence and analytics as a field spanning data, analytical methods, and the organizational use of information. In a logistics context, dashboards can make selected metrics accessible to users who need to compare performance across dimensions without repeatedly querying raw data.

More recent work has examined dashboards as a way to present supply-chain performance measures and support operational decisions. Vazquez-Noguerol, García-Arca, and Prado-Prado (2025) discuss a dashboard-based approach to road-freight performance measurement in a food-distribution company, emphasizing KPIs as a component of operational performance management. Their case differs from the present project in sector, setting, and implementation, so it is used as related context rather than direct evidence for the findings reported here.

Across this literature, three design principles are relevant. First, metrics should be clearly defined so users understand what each number measures. Second, visual comparisons should support investigation rather than imply causation. Third, data quality and context must be considered before decisions are made. This project applies those principles through explicit KPI definitions, separate treatment of delivery status and risk labels, and a limitations section. The paper does not claim that its dashboard is novel as a visualization technology; its value lies in applying a documented KPI framework to the supplied dataset and reporting the resulting descriptive patterns.

3. Dataset and Research Methodology

The analysis used a cleaned dataset containing 180,517 order records. The working dataset included fields for actual shipping days, scheduled shipping days, delivery status, late-delivery-risk flag, shipping mode, customer segment, order region, market, and other order/customer/product attributes. The source and collection process for the underlying data were not independently verified; therefore, the findings should be interpreted as results for the dataset analyzed, not as an audited assessment of APL Logistics.

The cleaning workflow removed 11 rows during preparation and removed two duplicate records. The final analytical table contained 180,517 records. The project's data-quality review reported no missing values in the final dataset and no negative values in the shipping-duration fields. These checks support the calculations used here but do not guarantee that every source value is semantically correct.

The principal calculated field was Delay Gap, defined as actual shipping days minus scheduled shipping days. A positive value indicates that actual shipping duration exceeded the scheduled duration; zero indicates equal duration; a negative value indicates shipping faster than scheduled. The observed Delay Gap values ranged from −2 to 4 days. Delivery Class grouped records into Delayed, Early, and On Time categories using the project's delivery-status logic. Delay Severity was defined as No Delay for Delay Gap ≤ 0, Low for values up to 2, Medium for values up to 5, and High for values above 5. Under this rule, no records fell into the High category.

The analysis was descriptive and cross-sectional. It used counts, proportions, averages, and comparisons across shipping mode, customer segment, region, and market. The dashboard was built in Microsoft Power BI, with measures and calculated fields used to summarize the data. No machine-learning model, hypothesis test, or causal identification strategy was applied. Consequently, the results describe patterns in the observed records and should not be interpreted as forecasts or causal effects.

4. KPI Framework and Dashboard Design

Four dashboard views organize the analysis: Delivery Performance Overview; Delay Risk Analysis; Shipping Mode Comparison; and Regional and Market Diagnostics. The views are intended to support a sequence from overall performance to focused comparisons.

The main KPIs are defined as follows:

• Total Orders: number of records included in the analytical dataset.

• On-Time Delivery Rate: share of records classified as On Time.

• Average Delay (+): average positive Delay Gap among delayed records, as represented by the project's measure.

• Late Delivery Risk %: share of records with late_delivery_risk = 1.

• Shipping Mode Efficiency Index: scheduled shipping days divided by actual shipping days, multiplied by 100, as a project-specific comparison measure. It is not a universally standardized logistics KPI. Where scheduled duration is zero, the ratio is undefined and should be displayed as not applicable rather than forced to a numeric value.

• SLA Compliance %: share of orders in a shipping-mode group classified as Early or On Time.

• Regional Delay Index and Market Delay Index: average positive Delay Gap for the selected region or market, using the project's calculation.

The dashboard also compares delivery class by shipping mode, late-delivery risk by customer segment, and scheduled versus actual shipping duration. Filters allow users to narrow relevant views by shipping mode, region/market, and customer segment. Because the dataset used for this project does not contain a usable date field, the analysis does not support a date-range slicer or time-series trend analysis.

A key methodological distinction is maintained between Delivery Class and Late Delivery Risk. Delivery Class is based on observed delivery outcome/status, whereas Late Delivery Risk is a separate label in the dataset. The two are not equivalent and should not be substituted for one another. For example, 103,399 records were classified as Delayed, while 98,976 records had late_delivery_risk = 1.

5. Results

The dataset contained 180,517 order records. The On-Time Delivery Rate was 42.72%. Delivery classification showed 103,399 delayed records (57.28%), 43,365 early records (24.02%), and 33,753 on-time records (18.70%). The delayed share was therefore greater than half of all records, while fewer than one in five records were classified as on time under the project's status grouping.

The late-delivery-risk flag was positive for 98,976 records (54.83%) and negative for 81,541 records (45.17%). This percentage is different from the delayed-order share. The difference is important for interpretation: the dataset contains 4,423 records classified as delayed that have a risk flag of zero. The risk flag should therefore be analyzed as a separate field, not as a replacement for the delivery outcome.

Shipping-mode comparisons revealed variation in the risk flag. First Class had the highest reported late-delivery-risk rate at 95.32%, followed by Second Class at 76.63%, Same Day at 45.74%, and Standard Class at 38.07%. These are descriptive associations in this dataset. They do not establish that a shipping mode causes higher risk; mode selection may be related to order characteristics, routes, service commitments, or other factors not evaluated here.

Scheduled and actual shipping durations also differed by mode. For Standard Class, the displayed averages were 4 scheduled days and 4 actual days. For Second Class, the scheduled average was 2 days and the actual average was approximately 3.99 days. For First Class, the scheduled average was 1 day and the actual average was 2 days. For Same Day, the scheduled average was 0 days and the actual average was approximately 0.48 days. These values suggest that the gap between planned and observed duration varies across service categories. The Same Day ratio-based efficiency index is undefined because its scheduled duration is zero.

The project's Shipping Mode Efficiency Index values were 100.0 for Standard Class, approximately 50.1 for Second Class, and approximately 50.0 for First Class. Same Day was not assigned a ratio value. SLA Compliance was reported as 60.23% for Standard Class, 52.17% for Same Day, 20.27% for Second Class, and 0% for First Class. Since the efficiency index is a custom ratio and SLA compliance depends on the delivery-class definition, the metrics should be read together rather than treated as interchangeable rankings.

The Regional Delay Index highlighted several areas with higher average positive delay gaps. Central Asia had a reported index of 1.68 days, Eastern Asia 1.65 days, West of USA 1.64 days, South of USA 1.64 days, and Northern Europe 1.63 days. These are differences in average positive delay gap, not percentages of late deliveries. They indicate where further investigation could be prioritized, but they do not identify the operational

causes behind the differences.

Delay Severity categories contained 77,118 records in No Delay, 89,364 in Low, and 14,035 in Medium, with no records classified as High under the project's threshold rule. Because this classification is based on Delay Gap thresholds and differs from delivery class, users should review the definition before comparing its counts with the Delayed category.

6. Discussion

The results point to three practical observations. First, the delivery-class distribution shows a substantial delayed share in the analyzed data. A service or operations team could use this finding to prioritize deeper investigation, but should first validate the dataset's provenance, period, and operational definitions before applying the result to a real company.

Second, shipping-mode risk rates vary considerably. First Class has a high positive-risk share in this dataset, while Standard Class has a lower one. This should not be interpreted as evidence that First Class service is inherently worse. A fair comparison would need to account for route distance, order size, origin and destination, customer expectations, service-level commitments, and the way shipping modes are assigned. A mode with stricter promised delivery times may show more failures even when its absolute transit time is shorter.

Third, regional indices may help narrow the scope of operational review. Higher average positive delay gaps in Central Asia and Eastern Asia could motivate route-level analysis, carrier review, customs or handoff checks, and validation of scheduled transit assumptions. These are possible investigative directions, not findings established by this dataset. The present analysis does not include event-level timestamps, carrier-level identifiers, weather, customs events, warehouse processing times, or other variables needed to isolate causes.

The dashboard's value is therefore primarily in visibility and prioritization. KPI cards summarize overall outcomes; shipping-mode comparisons show variation; and regional diagnostics identify groups for closer review. The interactive design helps users explore these patterns without treating the charts as proof of causality. To support operational decisions, the dashboard should be paired with subject-matter review and verified data definitions.

7. Limitations

Several limitations constrain interpretation. First, the dataset's original source, extraction date, sampling method, and relationship to actual APL Logistics operations have not been independently verified. The case-study label describes the project context and should not be read as proof of corporate ownership or endorsement.

Second, the dataset does not provide a usable date field for this project. The analysis cannot show changes over time, seasonality, before-and-after effects, or whether performance is improving or deteriorating. Third, the study is descriptive: it does not estimate statistical significance, predict future delays, or establish causal relationships.

Fourth, the risk flag and delivery status are separate variables. They differ in count and meaning, and users must avoid interpreting the risk percentage as the delayed-order percentage. Fifth, the Shipping Mode Efficiency Index is a project-specific ratio of scheduled to actual shipping days. It is sensitive to zero scheduled duration and is not directly comparable to standardized industry benchmarks without further validation. Sixth, aggregated regional results can hide variation among individual routes, countries, carriers, and order types.

Finally, the cleaning checks described here are limited to the steps recorded in the project workflow. Further validation should examine outliers, field semantics, consistency of delivery-status labels, the calculation logic behind the original risk flag, and whether records are independent. These checks would improve confidence before the analysis is used for operational or academic claims beyond this exploratory case study.

8. Conclusion and Future Scope

This paper documented a dashboard-based analysis of delivery performance, late-delivery risk, shipping-mode differences, and regional delay patterns in a cleaned dataset of 180,517 records. The overall on-time delivery rate was 42.72%, while 57.28% of records were classified as delayed. The late-delivery-risk flag was positive for 54.83% of records, confirming the need to keep risk labels separate from observed delivery outcomes. First Class had the highest positive-risk rate among the shipping modes displayed, and Central Asia had the highest reported Regional Delay Index among the regions listed.

The findings are descriptive and should be treated as a starting point for investigation rather than a verdict on a company, service, or transport option. A Power BI dashboard can help organize operational data into accessible measures and comparisons, but reliable decisions require verified data provenance, well-defined metrics, and context about how shipments are planned and executed.

Future work should add a reliable order or shipment date to enable time-series analysis; validate the source and business definitions; include route, carrier, origin/destination, and milestone data; and conduct statistical analysis of factors associated with delays. With adequate labeled historical data, a separately validated predictive model could be explored. Any future intervention should be evaluated with an appropriate before-and-after or quasi-experimental design rather than inferred from dashboard correlations alone.

References

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